Bahrain University
College of Engineering
Electrical and Electronic Engineering Department

Smart Street Lights

A Senior Design Project (EENG 490) Report

Submitted in partial fulfillment of the requirements for the degree of

B.Sc. in Electrical Engineering

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Abstract

In Bahrain, time-control and light-detecting methods are being utilized to control the condition of a streetlamp, meaning that their working precision is bad since environment-sensing and motion-sensing are absent. Along these lines, engineers are reevaluating this working standard to discover elective arrangements that meet the smart road lighting necessity at a lower cost.

The idea of the project is to build up an insightful unique lighting system that improves the present working guideline of streetlights, which is either two conditions "ON" and "OFF", with no detecting to the general condition changes or vehicle moving to find a solution for the depletion of energy resources.

This idea prevents the lights from working unnecessary, the instance of no vehicles or people on foot going past. The project utilizes an Arduino controller asset with a motion-detection sensor to wake-up the streetlights based on the traffic. Additionally, we will use LEDs rather than High-pressure sodium lights and look at the consumption difference between both by comparison. Moreover, a user interface is linked directly with the system for monitoring.

The implementation of an intelligent street lighting system does not just ensure energy-saving, but it can also help to extend the service time of the lighting itself since they are not turned ON permanently. Moreover, it spares the work expenses of maintenance and materials significantly.

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Chapter 1
Introduction

1.1 Motivation

During the nighttime, there are some roads in Bahrain where there is little traffic or pedestrian movement, especially late in the dead of night or early morning. the most problem is that the current lighting system relies only on time-control and light-sensing. In addition, the lights will activate supported a schedule, or by the assistance of Cadmium Supplied Photoresistors (CDS). “When many lights fall on that cell, then the light will be turned off, on other way if there are no enough lights to conduct the light will turned on” [1]. But this method is inefficient since the lights are working only within the “ON” and “OFF” state with no sensing to the encircling. as an example, if at a particular time there is no traffic or pedestrian passing, why do we need these lights?

By sensing the environment and the traffic state, we are able to implement a system that the sunshine intensity is often adjusted accordingly. And it often further improved to be applied across Bahrain lighting system as a component of the smart lights, which is that the hottest topic currently discussed by many who puts such a lot stress on the purpose of unpolluted, renewable and efficient energy consumption.

1.2 Objective

Our main objective in this project, is implementing and developing an intelligent streetlight system. this technique should be ready to get real-time readings from the sensors to sense the traffic condition within the road and adjust the light intensity accordingly. this may only be achieved and proved t by doing the subsequent steps:

1. Research and get the necessary data and information about the project.
2. Design a prototype to simulate a real-life scenario.
3. Programming the controller (Arduino).
4. Determine power consumption of lights used in Bahrain and LED that we are going to use in our project.
5. Comparing power consumption and the cost between the normal system used and the intelligent system that we are going to use.

1.3 Difficulties and constraints:

In our project have facet many difficulties that effect our works, some of them have overcome and others that we have not been able to solve it.

Some of the constraints that were solved are:

a) General situation and atmosphere

We were affected by the general situation and atmosphere due to the outbreak of the Corona virus since universities, schools and shops stopped working during this period, then we have faced troubles and limited access to the university lab to do some works like soldering, cutting some items by machine, and we do not have efficient power supply to operate the project probably.

b) Buying the items:

All of the gadgets had been bought online from "AliExpress and eBay" sites because it is priced very excessively if offered from Bahrain specifically when these objects cannot be found everywhere the only limited number of markets while all markets already stopped working at this period, therefore the disadvantage of the online ordering is the time of the shipping up to two weeks.

c) Designing the prototype:

We have used a big breadboard base on the project while that is only available with us, because we are not able to solder with small perf board due to limitation of lab access, therefore this is very big for designing a small system with a small area.

d) Time of work:

As we are senior students, we had 5-6 courses including the Senior project, so they had to study and focus on our courses and work on the senior in parallel especially in this semester with online studies, we face many difficulties and troubles and a lot of housework in it. Time played a huge factor in our progress in this senior.

1.4 Report Outline:

In the subsequent chapters, a small background, information, and some calculations will be presented and discussed. Also, we will be giving a short introduction to the layout and implementation of our project and how we have finished our objectives. Finally, the results and operation are also proven with future work noted as well.

Chapter 2
Background

As inception, it had been necessary to start with a small briefing about the current lighting system used in our country.

2.1 Street lighting poles

A streetlight, light pole, a raised source of light on the edge of a road or path. Modern lamps may also have light-sensitive photocells that activate automatically the sunlight take off. This function in older lighting systems could have been developed with new one. [2]

The types of the light poles depend on how many arms each light pole have, it might be single arm, double, or even four to six arms and more kinds that are being innovated to have better light distribution based on the application.



Figure 2-1: Single, Double & 4 arms light poles

The main problem that the current lighting system have in Bahrain is the way those lights are operating, which is either "ON" or "OFF". This principle works fine, but not good enough when it comes to energy consumption.

Therefore, streetlamps are orange because they are cheaper that way. The numerous disadvantages to people's safety are the main reason why many towns and cities have been moving towards LEDs and high-intensity discharge lighting that provide a cleaner white light - new technologies mean white light does not have to break the bank, and it could save a life or two." [3]

2.2 LED vs High Pressure Sodium (HPS)

Numerous municipalities are measuring the advantages of both LEDs and high-pressure sodium (HPS) as they are reconsidering their road lighting choices since LEDs have proved their efficiency lately. LEDs are a better choice relative to most of the lighting options currently on the market since they deliver better lighting after all losses are accounted. On the other side, HPS lights performs very poorly when compared to LEDs

when it comes to lifetime cost, HPS lights have to be replaced regularly to maintain their brightness. However, individual LED diodes keep their brightness overtime. Regular HPS lights can last for approximately 10,000 hours while LEDs can last for 50,000 to 100,000 hours meaning that you will have longer operating period with better efficiency which means better use of the consumer's money.



Energy Efficiency	Low	High
Durability	Fragile – has moving parts, glass bulbs, and filaments	Heavy-duty, has no electrode or filament, shock & vibration resistant
Lifetime	10,000 to 25,000 hours	100,000 hours
Lumen Depreciation	Moderately high	Low
Operating Temperature	-30° to 65°	-55° to 70°
Performance	Requires 5-10 minute warm-up time, creates light in all directions	Turns on instantly, no flickering, creates focused light
Color Quality	Average	Superior
Light Pollution	High	No
Dimmable/controllable	No	Yes
Cost	Lower upfront cost, but requires regular relamping and ballast replacement	Higher upfront cost but virtually no maintenance expenses

Figure 2-2: LED vs High Pressure Sodium Lighting [7]

Another important point would be an environmental one. Lights in general produces heat, since lightbulbs get hot because the filament is being supplied by electrical energy and some of this energy is dissipated as heat rather than light. This concept applies for both LEDs and HPS, with LEDs being more efficient (produces less heat). That is important since we live in a country which is very hot during the summer, and with currently used lights they will produce more heat as well.

2.3 The current Bahrain street lighting Power Consumption

Regularly, the power consumption for the main roads differs based on the type of pole that is installed and for rural roads and internal roads. The operation losses for the different poles according to Electric and water authority is rated +20%. The lights which are distributed in Kingdom of Bahrain operate for approximately 4280 hours yearly.

$$Average\ of\ operation\ in\ year: \frac{4820}{360}\left(\frac{Hours}{days}\right) = 11.8\ hours$$

Figure 2-3: HPS vs LED [12]

As stated previously, the power consumption for each pole type is different as shown below.

Table 2-1: Power consumption for each pole type

Hight / type	Single arm	Double arm	Four arms
5 m	70 w	140 w	280 w
8 m	250 w	500 w	1000 w

2.3.1 Active Hours

Active or working hours of the lighting system are crucial since they mean more power consumption. So, in order to have more accurate results and efficient system, the operating hours must be set carefully.

2.3.2 Power calculations and comparison

For 12 hours of operation, the power consumption will be different depending on the height and rated power of each pole. The calculation of the power considers the active hours for the poles. The power of each kind of pole will be calculated based on the power equation shown below. By Assuming the losses is 0.2 in 12 active hours

$$Power\ Consumption\ (kWh) = Rated\ Power * Losses * Active\ Hours * No.Poles$$

A- <u>HIGH PRESSURE SODIUIM LIGHT:</u>

Table 2-2: Power consumption of HPS light for 5m pole

5 m pole		
Number of bulbs	**Rated power (W)**	**Power consumption (kilowatt per hour (kWh))**
1	70	3528
2	140	7056
4	280	14112

Table 2-3: Power consumption of HPS light for 8m pole

8 m pole		
Number of bulbs	**Rated power (W)**	**Power consumption (kilowatt per hour (kWh))**
1	250	12600
2	500	25200
4	1000	50400

B- **<u>LED LIGHTS:</u>**

$$LED\ Power = Rated\ power(W)\ of\ High\ Pressure\ Sodium * 0.65$$

Table 2-4: Power consumption of LED lights for 5m pole

5 m pole		
Number of bulbs	**LED Power**	**Power consumption (kilowatt per hour (kWh))**
1	45.5	2293.2
2	91	4586.4
4	182	9172.8

Table 2-5: Power consumption of LED lights for 8m pole

8 m pole		
Number of bulbs	**LED Power**	**Power consumption (kilowatt per hour (kWh))**
1	162.5	8190
2	325	16380
4	650	32760

From the tables above it is clearly seen that the power reduced by 35% with a huge gap or amount of power saved in each pole with different bulbs meaning that the LEDs are more efficient solution compared to HPS, and this will defiantly reflect on the cost which will be discussed in next chapters.

Chapter 3

Requirements

First, we had to choose which devices and programming tools that we are going to put into effect to achieve our tasks.

3.1 Arduino

All electronic and electrical devices are connected to the controller to execute specific operations. The controller is the brain of each device. Different devices require different types of controllers. Nowadays, there are many controllers such as Arduino and Raspberry Pi. Choosing the controller depends on the types of specific operation of the project and the number of inputs and outputs that are contained. In addition, the size of memory is very important and that is why Arduino is the perfect choice for this project.

3.1.1 Why Arduino?

Inexpensive – comparing the Arduino boards to other microcontroller platforms. Arduino boards take the advantage since they are relatively inexpensive. Also, the company provides boards that can be assembled by hand as they are the least expensive version of the Arduino, but even the Arduino boards which are pre-assembled they cost less than $45. [4]

Cross-platform - comparing the Arduino boards to other microcontroller platforms from the durability wise, the Arduino Software (IDE) runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows XP.

A familiar programming environment - The Arduino Software (IDE) is flexible enough for students, Teachers and advanced programmers. As it does not cross the comfort zone for all users since it is based on the regular programming languages C/C++, which is it known for most people.

Open source and extensible software - The Arduino software is published as open source tools, and it is allowed to be extended by experienced programmers. The language can be expanded through regular C++ libraries, and people wanting to understand the technical details can make the leap from Arduino to the AVR C programming language on which it's based.

3.1.2 Specifications of Arduino

Arduino project is considered to be as an open-source hardware and software for the user, Arduino company was licensed under the GNU Lesser General Public License (LGPL) which allows the products of Arduino boards and software to be used and distributed to anyone.

Arduino boards provide both digital and analog input/output pins that can connect and interfaced with so many expansion boards and circuits and also communications interfaces such as the Universal Serial Bus (USB) on the board, so you can upload your programs and communicate with the board.

On the Arduino boards microcontroller which is mainly programmed using a regular programming language such as C and C++ which provide easy access to this world since those languages are famous and easy to be learned. In addition to using traditional compiler languages, the Arduino boards provide an integrated development environment (IDE) based on the Processing language project.

As Arduino uses standard connectors - which gives the board a great advantage- since it enables users to connect the CPU board to a variety of interchangeable add-on modules such as: sensors, receivers, shields any many others.



Figure 3-1: Some of Arduino modules

The Arduino board can be powered by three power supply options the 5V USB, DC jack and Vin(pin). The DC jack requires 7V to 12V input, and it will draw a maximum current of 900ma. The Vin pin has the same voltage level as the DC jack draws the same amount of current. On the hand, the USB input requires 5V and draws a maximum current of 500ma.

For the sake of our project, we have chosen the Arduino Mega over Arduino Uno since it contains a large number of digital inputs and output (54 pins) and 14 pins can be used as PWM. In addition, it has the 16 input analogue pins.

Table 3-1: Arduino UNO [8]

UNO	
Microcontroller	ATmega328
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limits)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	40 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (ATmega328) of which 0.5 KB used by bootloader
SRAM	2 KB (ATmega328)
EEPROM	1 KB (ATmega328)
Clock Speed	16 MHz

3.2 Motion Sensor

3.2.1 PIR

Motion sensors, or sometime called Passive Infrared Sensors (PIR) since it is based on infrared technology, is a high sensitivity, high reliability, ultra-low-voltage device that can be used to detect the movement of objects in or out of his region. They are widely used since they are inexpensive, cheap and easy to use.

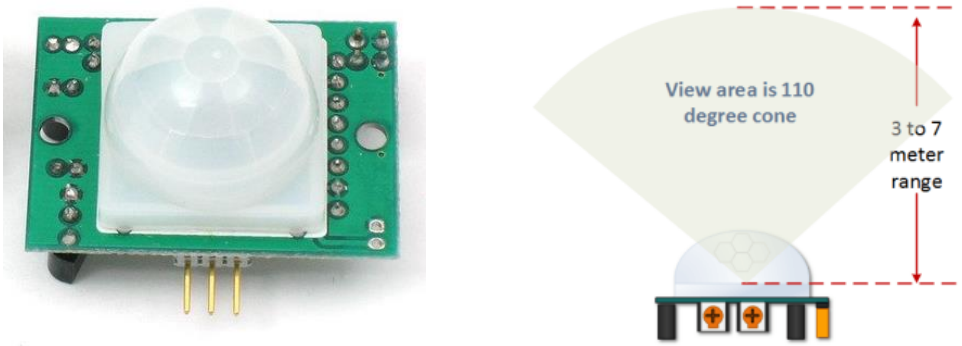


Figure 3-2: Motion Sensor

The motion sensor or PIR has a pyroelectric sensor inside, after removing the plastic dome. Pyroelectric sensor is an infrared sensitive optoelectronic component which are specifically used for detecting electromagnetic radiation in a wavelength range from (2 to 14) µm. Everything emits some level of radiation, and the hotter the object the more radiation will be emitted. Unlike semiconductor detectors, a pyroelectric detector is a thermal detector working with a thermally isolated chip which is covered by a black absorption coating. This coating converts the infrared radiation falling on the chip to heat. The chip changes its temperature in the order of magnitude of µK … mK and, as a result of the pyroelectric effect, produces the electrical signal desired. [5]

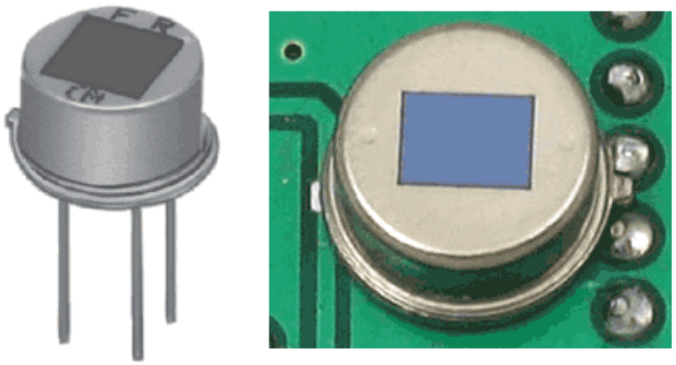


Figure 3-3: Pyroelectric Sensor

3.2.2 How does it work

The PIR sensor itself is split into two slots; each slot is made of a special material that is sensitive to IR. The reason why this is split into two halves is to detect change of

motion not average IR levels. When the sensor is in the idle state, both slots will be reading same IR percentage, indicating no change of motion. When an object passes by the PIR range of detection, it intercepts one slot of the PIR sensor indicating a change of motion and vice versa when an object leaves. In other words, the two halves are connected together and when one of them reads more than the other, the output will be either high or low.

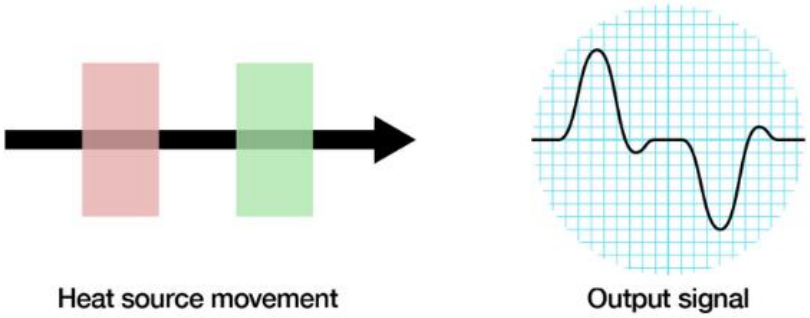


Figure 3-4: Heat movement and output signal relation

The lens that is covering the internals of the sensor does more than what is seems to be doing. The way that the lens is crafted can manipulate the breadth, range, sensing pattern very easily. PIR sensors use Fresnel lens, this lens condenses light providing a larger range or IR to the sensor.

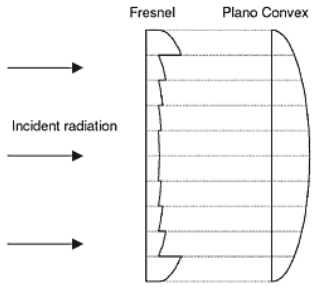


Figure 3-5: PIR Sensor lens structure [6]

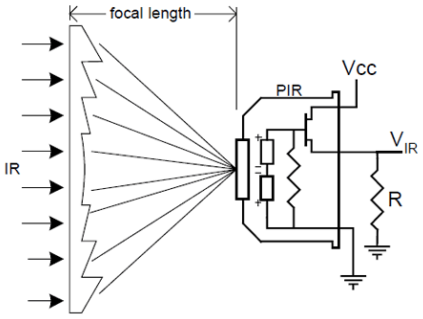


Figure 3-6: Fresnel lens condensing IR [6]

Hence, thanks to Fresnel lens we can have a wider range of detection. However, remember that practically we actually have two slots as mentioned previously, and more importantly we do not want uncontrollable/random big sensing-area, but rather a

scattering of multiple small areas. So, the lens is somehow divided into multiple sections, each section, or as they call them “sub-lenses”, represents a Fresnel lens itself.

3.3 Java

Java is basically a computer programming language, like C, C++, Python and other languages. But what is interesting about java is that it allows you to program with instruction set much easier compared to other programming languages. In other words, it uses commands which are “English-based” making it very easier to deal with. However, you cannot just write whatever you like in English thinking that Java will understand what you are saying since similar to all programing languages there is a syntax which you have to follow.

3.3.1 Why Java?

Easy to Use – Java is based on C++, however the syntax of C++ can be complex and confusing. Java uses fluent English-like syntax making it easy to real and learn.

Free – Money plays a huge factor everywhere. If people are to pay money for learning a language, most of them would rather not. Since Java is free, and also powerful, then that gives it a huge benefit over other competitors.

Platform Independent – Once you write your program using Java, then it can run on any operating system or hardware you would imagine.



Figure 3-7: Java logo

3.3.2 Johnny-Five

JavaScript is all over the place: running applications, running the servers that sustain HTML to the program, and running databases, to give some examples. It ought to be nothing unexpected, then, that JavaScript can likewise be utilized to control off-the-rack equipment, circuits and different hardware like the Arduino. Using systems like Rick Waldron's Johnny-Five for NodeJS, and the "StandardFirmata" programming bundle for Arduino, you can compose JavaScript on your PC and utilize it to control and control almost any part of an Arduino-based equipment setup.

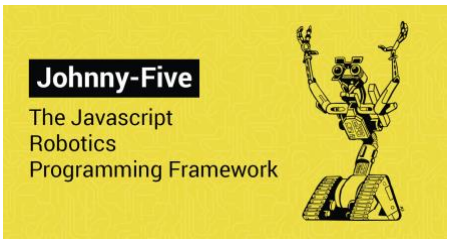


Figure 3-8: Johnny-Five

Chapter 4

Design and Implementation of prototype

This chapter discusses each of the design phases of the project from many aspects until the project objectives are reached. Each section will discuss a phase and we will have a detailed look at each one.

The diagram below shows the initial block diagram used in the proposal of this project.

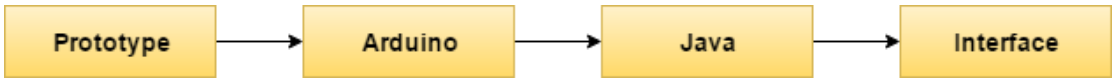


Figure 4-1:Block diagram of design phases of the project

4.1 Actual Road design

Since we are going to simulate a real-life scenario based on a road which already exist, King Hamad highway to be more specific, it was necessary to design a prototype to get a closer idea about how we are going to implement our circuitry later.

As a base for our project, the have bought 2 meters×1-meter foam board instead of plywood which is a heavier material. Foam board is usually lightweight, yet strong and cuts easily which makes it ideal to deal with. We have covered it in half so it do not break into halves when carrying around.



Figure 4-2: Foam board used for a base

The prototype will be as the road for our project, we have chosen it to be 6-lanes highway to replicate the biggest scenario that is out there. The prototype dimensions were taken to-scale when compared with the actual road measurements, the dimensions of the prototype are shown below.

4.2 Light poles

For the light poles, there were a lot of options ahead of us. Eventually, we have used plastic wire covers made from Bahrain-based Gulf Aluminum Rolling Mill (GARMCO). Plastic wire covers are better for wire management since they can be opened from one side which made the wiring hassle a lot better for use. Furthermore, they are easy to bend and drill. Those plastic covers were customized based on our need.

First, the plastic must be bended and then glued to form the arm of the pole. Then, the piece which will hold the LED circuit should be attached as well as drilling holes for the LEDs to pass from.

After forming the poles, which was a long process since this has to be repeated 16 times since we have 16 poles in our design, the poles are then painted and assembled together.

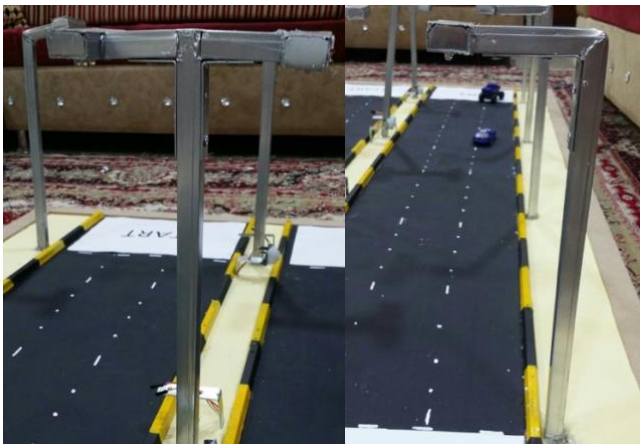


Figure 4-3: Two-arms & single-arm poles final result

4.3 Motion sensors position

Motion sensors are a vital part of our project, the motion sensors have been positioned in the prototype using L-angles (so we can adjust the angle of view easily) across the median. A total of 8 motions sensors are positioned, 4 for each side of the highway for the system to be more accurate and controllable.

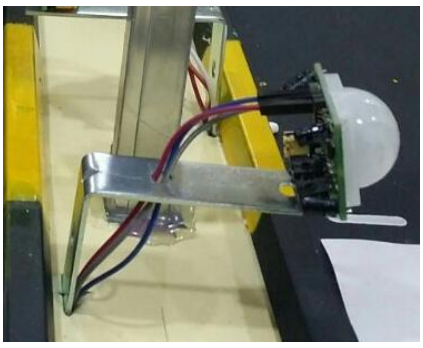


Figure 4-4: Motion sensor positioned

4.4 Control unit

For the purpose of our project, we will have 2 modes of operation: First mode which is always on, so the drivers can see what is ahead of them. The second mode is the important one, which is the mode that is going to operate when the car is in the motion sensor range. According to fuzzy controlling, these modes can be more specific and can take another possibility.

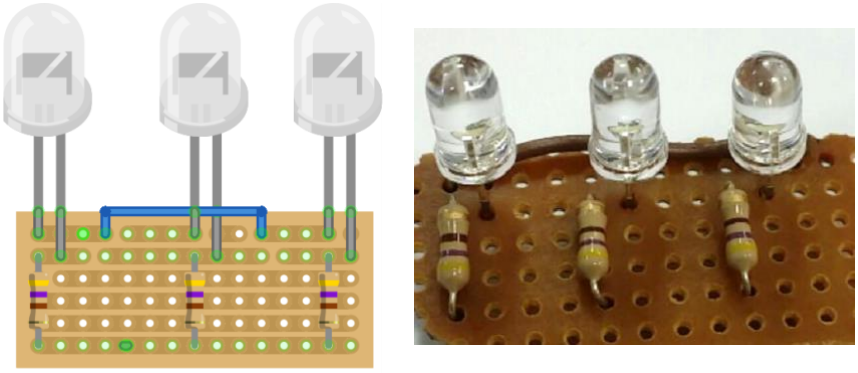


Figure 4-5: LEDs circuit on stripboard

For the LEDs circuit shown in the figure above, the anode side is connected with a 470Ω and the cathode is connected to the MOSFET. The MOSFET works as switch for the negative side, and will also help reduce the flickering and guarantee a constant voltage supply to the LEDs. MOSFET's gate is connected to the Arduino as a switch (ON/OFF) depending on the output of the Arduino based on the sensor's reading. The drain of the MOSFET is connected to the cathode LEDs, while the source is grounded. All of the LEDs are supplied from the Arduino with 5V to have brighter lighting so that the illumination is better for the drivers to see what is ahead. Of the 3 LEDs, the two which are on the sides are connected together using a jumper, which will work as a separate mode, while the middle one also separated.

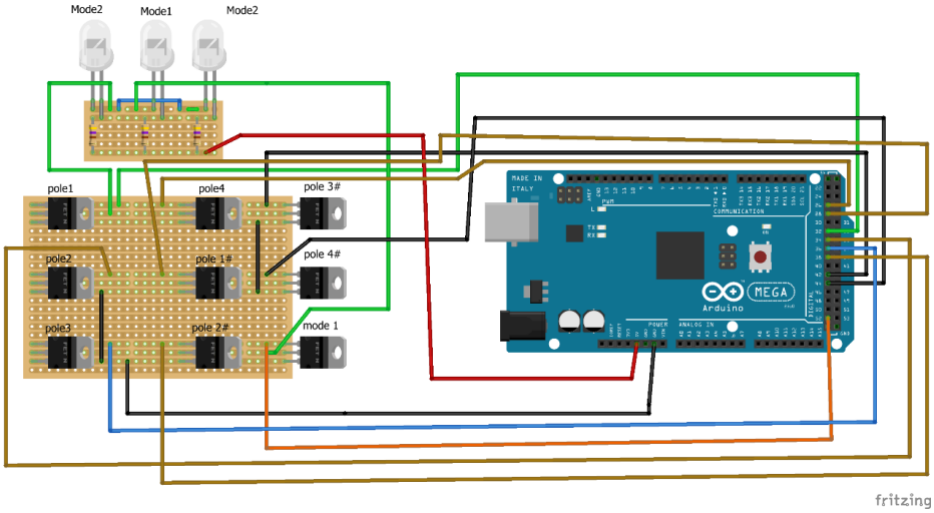


Figure 4-6: Control unit circuit using Fritzing (without motion sensor)

4.5 Motion Sensor

The design of the project requires using eight motion-sensors in total, four for each side of the highway, to detect any movement in the motion sensors' range. The motion sensor can measure from 3 meters to 7 meters. The motion sensor we have used is HC-SR501. This motion sensor module uses the LHI778 Passive Infrared Sensor and the BISS0001 IC to control how motion is detected. The module features adjustable sensitivity that allows for a motion detection range and also includes time delay adjustments and trigger selection that allow for fine tuning within your application.

HC-SR501 Pinouts and controls:

The pinouts and control of this specific motion sensor are shown in the figure below:

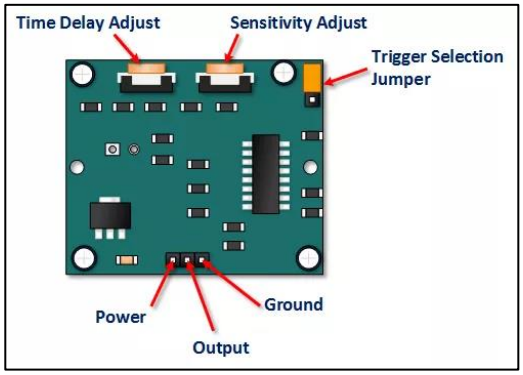


Figure 4-7: HR-SR501 Pinouts and controls [8]

The motion sensor is connected to the Arduino as shown in the figure below completing the circuit of the design. In the circuit diagram only four sensors are shown while actually there are eight of them. Same goes for the LED circuit, what is shown is one while there are 16 of them.

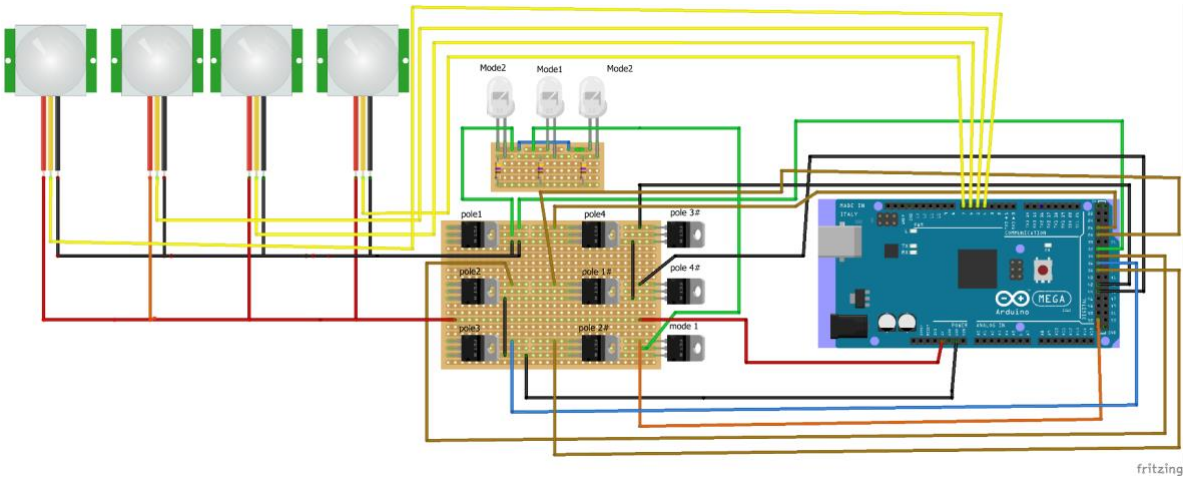


Figure 4-8: Complete Circuit diagram using Fritzing

4.6 LDR

As the theoretical implies, Light Dependent Resistor (LDR) is a light identifier which is utilized as a part of our circuit. LDR is basically a resistor, and its resistance changes as per the measure of light falling on its surface. At the point when LDR recognizes light, its resistance will get diminished, accordingly, if no light is falling on its surface (dark), then its resistance will increase. We have used LDR since it can be affective but later a proper scheduling of day/night times can be applied.



Figure 4-9: Light Dependent Resistor (LDR)

Chapter 5

Results and discussion

5.1 Prototype simulation

We have programmed a system that is connected to a server that is capable of controlling system either automatically or manually by using bottoms that can switch "OFF" and "ON" each individual pole lighting. Also, each pole is can be controlled separately which can be beneficial during any problems, issues or even a maintenance if the motion sensors didn't detect movement. Moreover, the system will show LDR's reading wither it is "DARK" or "Light". LED column of the table describe the status of the LEDs in the poles which appears "ON" or "OFF" and the exact time of movement detection in each sensor and that's also used to calculate the power consumption of the LEDs with accurate and efficient way. Last thing of our control systems that appears in the screen through the server is the graphical user interface that describe and represent the real implementation of the project which it shows the status of each LEDs wither its ON or OFF and change due to the change of LEDs status. Therefore, we have tested our prototype to make sure that it is operating the way we wanted. For the sake of simplicity, we have used only one side of the prototype to illustrate the idea of the project:

First scenario:

The first scenario covered is the most basic one, a car entering the road and finshing without any issues or problems. We will be showing the results we have got on the prototype along with the results from the interface we have created.

1. The car has entered the road so the first motion sensor has detected and increased the counter. The first two poles ahead of the car are triggered (to increase its lighting) and the time of movement is recorded as well.

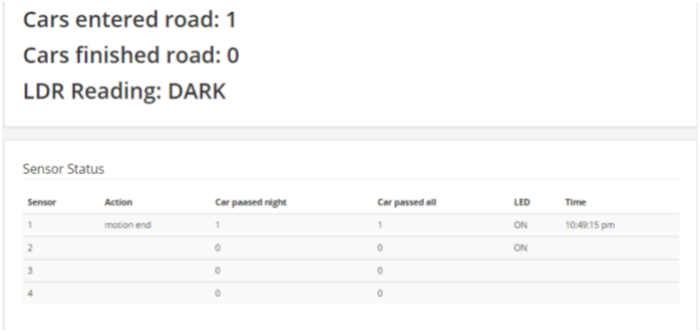


Figure 5-1: Counter increased, and time recorded in the interface



Figure 5-2: First two poles are triggered.



Figure 5-3: The car entering the road

2. The second is when the car comes closer to the second motion sensor, so the counter of sensor 2 will increase and pole 2 and 3 will turn on. When the counter of the second sensor equal to the counter of the first sensor that's means that the car moved from the first sensor (pole) to the next sensor and passed the first pole so no need to be ON so it will switch it off and will switch the third pole ON as shown in the table of the LEDs status or clearly shown in the interface that the first LEDs poles was switched OFF but it keeps mode 1 LED ON in order to be safety for drivers to see the street.

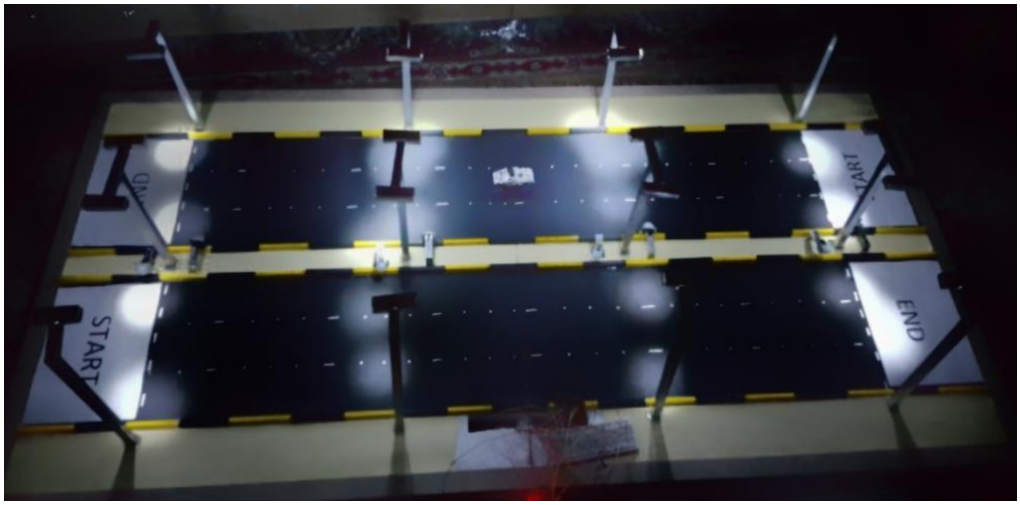


Figure 5-4: Car moved to next position

Sensor Status

Sensor	Action	Car paased night	Car passed all	LED	Time
1	motion end	1	1	OFF	10:49:15 pm
2	motion end	1	1	ON	10:49:27 pm
3		0	0	ON	
4		0	0		

Figure 5-5: Sensor status updated for pole 2

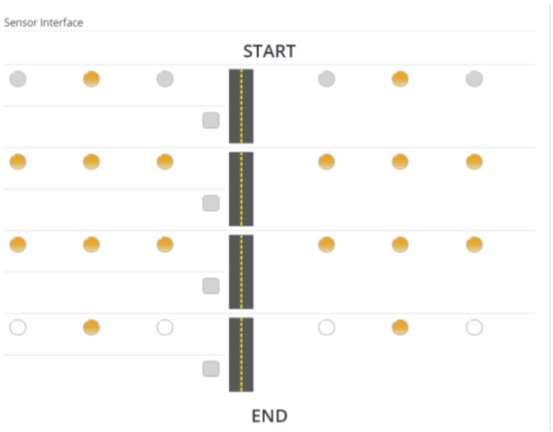


Figure 5-6: Pole 2 and 3 are triggered

3.The same concept will apply for the third pole (or sensor):

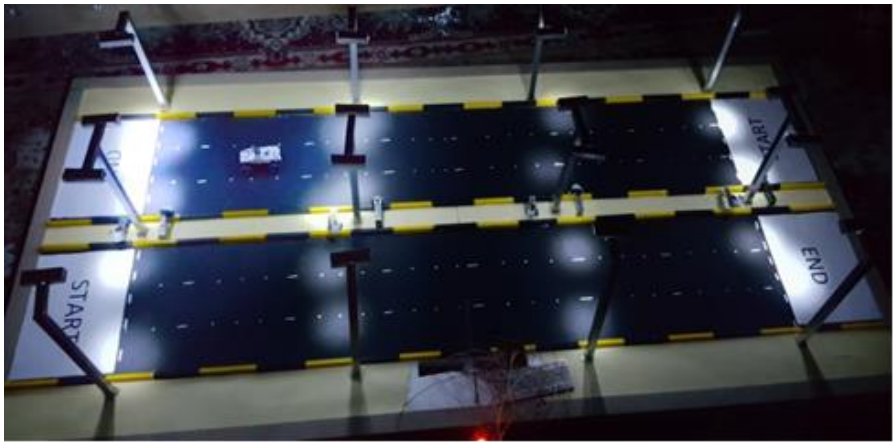


Figure 5-7: Car moved to third position

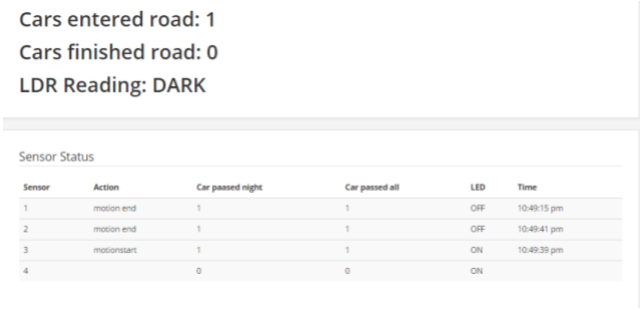


Figure 5-8: Sensor status table updated for position 3

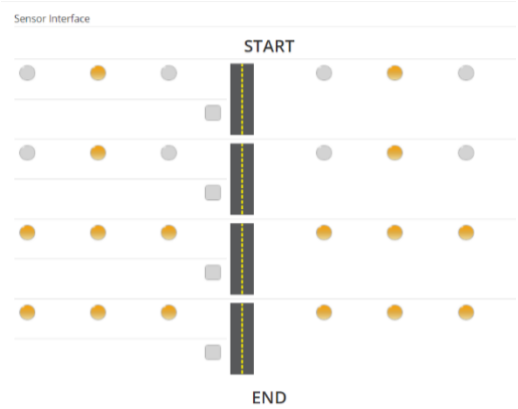


Figure 5-9: Third and fourth pole are working

4. When the car exits the road, which means passing the last motion sensor, the system will be able to recognize that the car left the path and show that on the interface with the exact time recorded. Also, it will check if counter of sensor 4 is equal to counter of sensor 3 that's means the car passed the third pole so no need to keep it operating, meaning there is no cars are behind this one. The last pole is programmed to work on a timer which will be delayed for 3-5 seconds which can be adjusted based on the need.

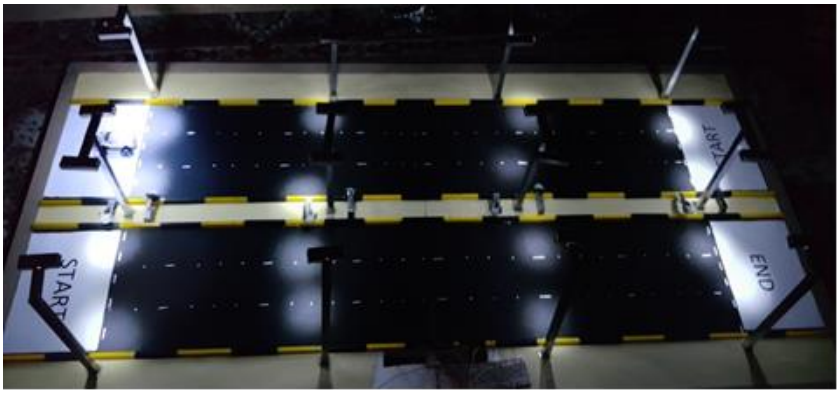


Figure 5-10: Car is in the last position

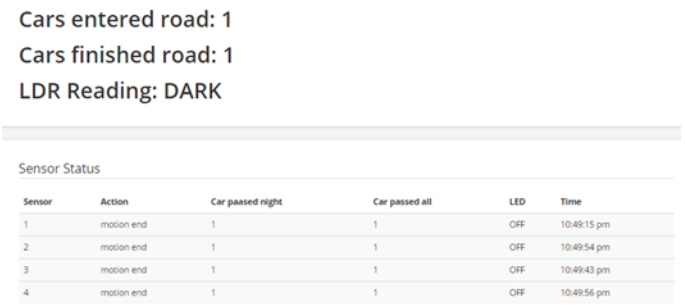


Figure 5-11: System indicates that car has left and status are updated.



Figure 5-12: Four pole is the only one operating and timer is triggered.



Figure 5-13: Road is clear, so last pole turned off

Second Scenario:

The scope of the second scenario discusses the unusual case. For example, a car stops or an accident happens which requires help. Keeping in mind that the lights will only turn on when there is car movement. The solution for such case is presented in the following steps below.

1. The first step is similar to scenario 1. One car will pass sensor 1 and will switch pole 1 and 2.

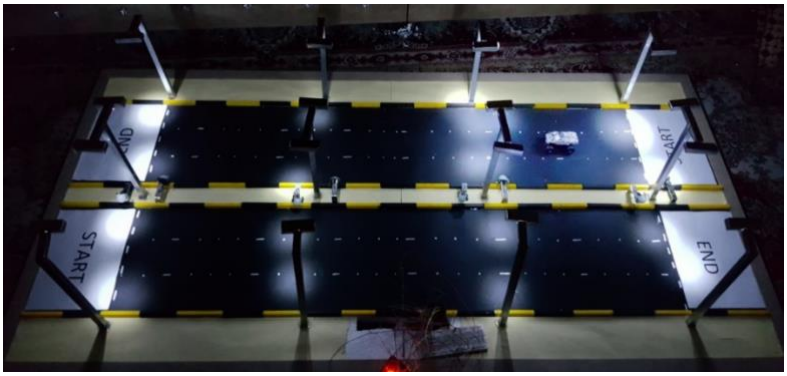


Figure 5-14: Car is in the first position

2. Also, the second step is same as before (second step of scenario 1) that the car passed second sensor and switched pole 1 OFF & pole 3 ON. But unfortunately, the first car has stopped in the middle of the street after passing the second sensor. Now, a second car has also entered the path, which means that first, second and third pole should be operating since there are two cars, one which has stopped between pole 2 & 3 and another one which just entered.



Figure 5-16: New car entered the road with and the previous stopped

Cars entered road: 2
Cars finished road: 0
LDR Reading: DARK

Sensor	Action	Car passed right	Car passed all	LED	Time
1	motion end	2	2	ON	7:25:36 pm
2	motion end	1	1	ON	7:25:12 pm
3		0	0	ON	
4		0	0		

Figure 5-15: First motion detects the new car



Figure 5-17: All the first three poles are operating

3. After that the second car passed the second sensor, the counter of each sensor will increase so it will switch off pole 1 again. But suddenly, a car accident occurs in the road segment between pole 2 and pole 3 and the two cars had to pull over. The interface shows that two poles where the accident happened at are operating due to this situation.

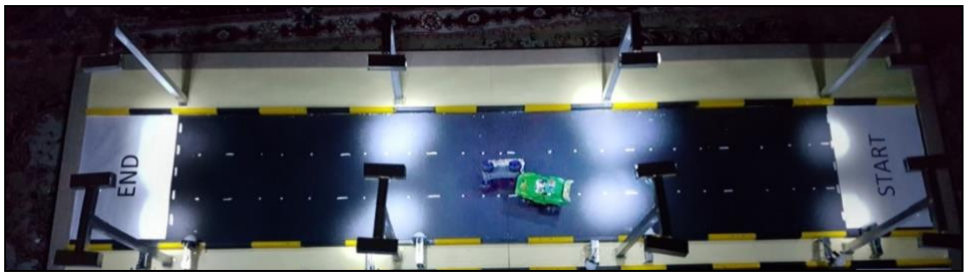


Figure 5-18: Car accident between pole 2 and 3



Figure 5-19: System shows that pole 2&3 are working

4. While the two cars are in a static condition at that road segment. A new car entered the road and the sensor detected that movement. Now the screen shown the number of cars which in this case is 3 cars. It will increase the counter of the first motion sensor so, again it will switch pole 1 ON as shown in the figure that three poles are operating.

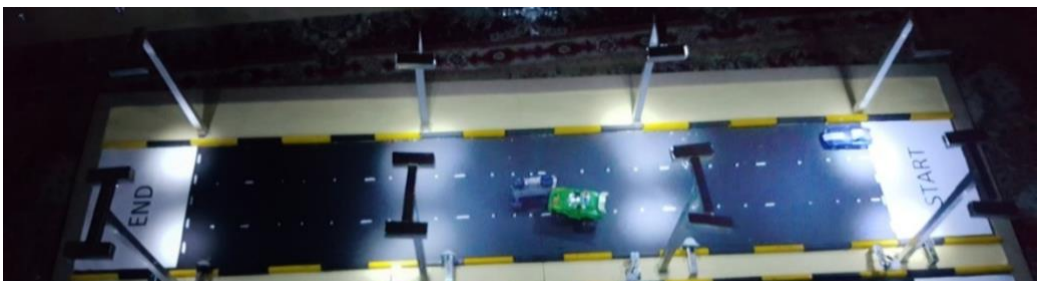


Figure 5-20: Third car entered the road

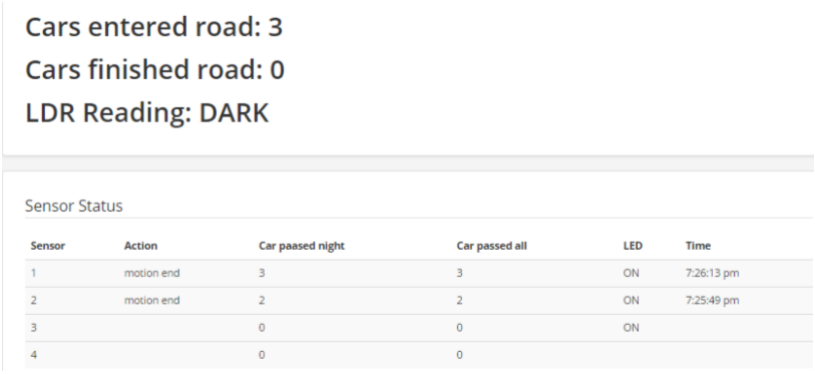


Figure 5-21: Third car entering the road

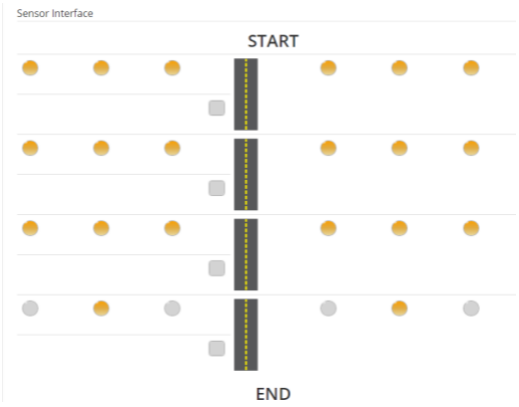


Figure 5-22: First three poles are working

5. The new car passed the second sensor and reached to the point of the accident but, if the driver is not in a position that allows him to stop.

Then the counter of sensor 1 is equal to the counter of sensor 2 so, it will switch pole 1 off again. As seen in the interface or the table showing the status of each LEDs in the poles.

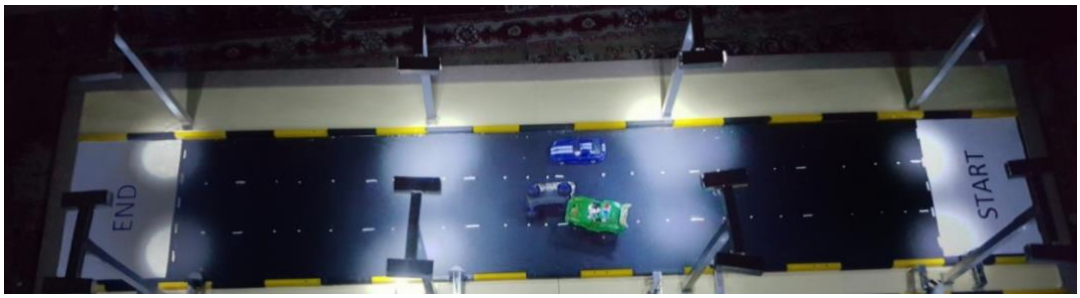


Figure 5-23: Third car passes normally



Figure 5-24: Second and third poles are working

6. When the new car passed the third sensor and the two previous cars are still in a static condition in this situation the motion sensors cannot sense the movement of the two cars or it is hard to detect a car while there is no movement. So, after passing the third motion sensor it will check the counter of each sensor and it will find that the counter of the third sensor is less than the counter of sensor 2 which means that there is an issue still in the road. According to this situation it will keep the poles near to the accident operating. As shown in the figure below the status of the three poles is ON and it can be seen in the interface clearly.



Figure 5-25: Car passed the accident to fourth pole

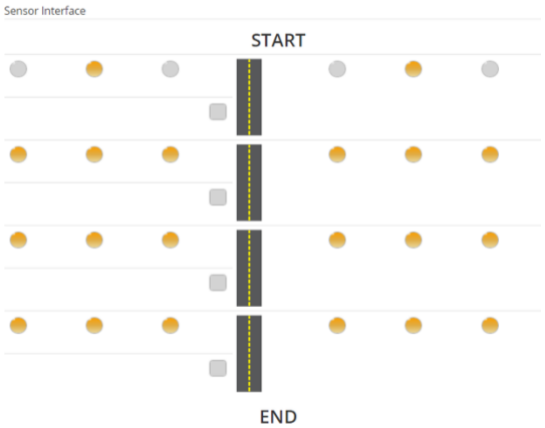


Figure 5-26: Last three poles are working

7. The car will finish the road with the accident still present so all the poles will shut-off expect the poles which are close to the accident.



Figure 5-28: Third car finished the path with the accident still present

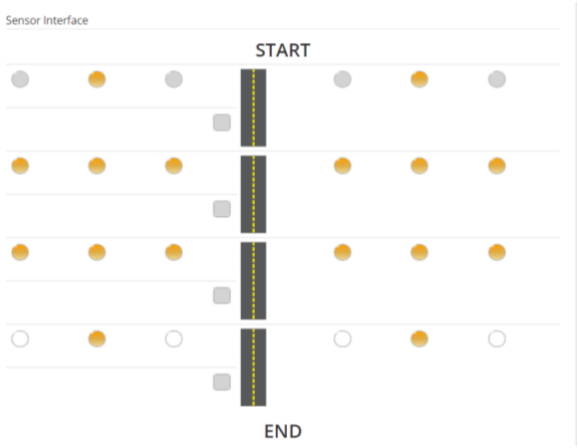


Figure 5-27: Poles which are close to the accident are still working

5.2 calculation and comparison

The assumption of this Calculation and comparison based on the information from Electricity and water authority. It shows the power consumption of each pole on 1 km and the comparison of the calculation related to the system applied and the intelligent one.

The intelligent system contains 2 modes of operation “MODE 1 and MODE 2” and they change the power rated of the LED in the intelligent system unlike the real one.

MODE1: the standard mode of lighting which is shows the road on regular situation for the upcoming vehicles and it makes the poles operates 30% from the rated one so it will save 70% of the power so this MODE for no motion on the road, so the following equation expresses the MODE 1

$$MODE\ 1 : 0.3 * LED\ rated\ power * Losses$$

MODE 2: the accomplishment of the operation with 100% of LED rated power and it works based on motion detection. Whenever the car enters the road the sensor is going to detect its motion and the rest of 70% of the light is going to be existed to give a sufficient level of lighting for the car passing through.

The total power consumption for the two modes will be

$$TOTAL\ POWER\ CONSUMPTION: Power\ consumption\ of\ MODE\ 1 + Power\ consumption\ of\ MODE\ 2$$

And we recall the power consumption equation mentioned above to calculate the total of the two modes.

These tables shows the MODE1 during 12 hours of operation and the cost for each type and the power consumption with the cost as it mentioned with the equation above.

Table 5-1: Mode #1 cost & power consumption for 12 hours of operation for 5m pole

5 m pole					
Bulbs	**Power Rated (W)**	**Active Hours (hr)**	**MODE 1 POWER**	**Power Consumption (kWh)**	**Cost (BD)**
1	45.5	12	13.65	687.96	19.95084
2	91	12	27.3	1375.92	39.90168
4	182	12	54.6	2751.84	79.80336

Table 5-2: Mode #1 cost & power consumption for 12 hours of operation for 8m pole

8 m pole					
Bulbs	**Power Rated (W)**	**Active Hours (hr)**	**MODE 1 POWER**	**Power Consumption (kWh)**	**Cost (BD)**
1	162.5	12	48.75	2457	71.253
2	325	12	97.5	4914	142.506
4	650	12	195	9828	285.012

As we did a study in King Hamad highway and the average number of cars was 7 cars so we assumed to put the average is 10 cars during 12 hours.

For the following calculation we assumed that the maximum speed, average and minimum and the distance is assumed to be on range of 1km, so the active time will be based on the speed.

$$Speed\ \left(\frac{km}{h}\right) = \frac{distance(km)}{time(hr)}$$

We **assumed** the total cars that will pass the road is 10 cars per hour.

Based on average Speed

The maximum speed we assumed is 90 km/h. so the time a car will take to cross the 1 km by this speed is 67 seconds. The following calculation of power consumption shows a car with 90 km/h. And the average of the active hours in this state is for MODE 1:

$$Average\ of\ Active\ Hours = (hours\ of\ MODE1(min.speed) + hours\ of\ MODE1\ (max.speed))/2$$

For MODE 2:

$$Average\ of\ Active\ Hours = (hours\ of\ MODE2(min.speed) + hours\ of\ MODE2\ (max.speed))/2$$

Table 5-3: Total cost & power consumption based on average speed for mode #2 (5m pole)

5 m pole							
bulbs	**Power Rated (W)**	**Active Hours (MODE1) (hr)**	**Active Hours (MODE1) (hr)**	**MODE 1 POWER (W)**	**MODE 2 POWER (W)**	**Total Power Consumption (kWh)**	**Cost (BD)**
1	45.5	9.75	2.25	13.65	45.5	988.9425	28.679
2	91	9.75	2.25	27.3	91	1977.885	57.358
4	182	9.75	2.25	54.6	182	3955.77	114.71

Table 5-4: Total cost & power consumption based on average speed for mode #2 (8m pole)

8 m pole							
bulbs	**Power Rated (W)**	**Active Hours (MODE1) (hr)**	**Active Hours (MODE1) (hr)**	**MODE 1 POWER (W)**	**MODE 2 POWER (W)**	**Total Power Consumption (kWh)**	**Cost (BD)**
1	162.5	9.75	2.25	48.75	162.5	3531.938	102.426
2	325	9.75	2.25	97.5	325	7063.875	204.852
4	650	9.75	2.25	195	650	14127.75	409.704

5.2.1 Comparison between (HPS) Intelligent LED System:

The comparison of the real system and intelligent system categorized based on the speed and calculation below for each type and the power saved with it cost.

The cost saved is the difference between the power consumption between the real and intelligent systems. The following tables show all the calculations.

5.2.1.1 Power comparison

Based on Average Speed

Table 5-5: Power saved by Intelligent LED system compared to HPS for 5m pole (average speed)

5 m pole			
bulbs	**HPS power consumption (kWh)**	**LED System Power consumption (kWh)**	**Power saved (kWh)**
1	3528	988.9425	2539.058
2	7056	1977.885	5078.115
4	14112	3955.77	10156.23

Table 5-6: Power saved by Intelligent LED system compared to HPS for 8m pole (average speed)

8 m pole			
bulbs	**HPS power consumption (kWh)**	**LED System Power consumption (kWh)**	**Power saved (kWh)**
1	12600	3531.938	9068.062
2	25200	7063.875	18136.13
4	50400	14127.75	36272.25

The gap between the intelligent and (HPS) for each type is obvious so to save power the intelligent one dominates in saving over the (HPS), even if the real one beats the LED System which operates 12 hours system.

5.2.1.2 Cost comparison

The comparison of the real system and intelligent system categorized based on the speed and calculation below for each type and the power saved with it cost.The cost saved is the difference between the cost between the real and intelligent systems. The following tables show all the calculations.

Based on Average Speed:

Table 5-7: Cost saved by Intelligent LED system compared to HPS for 5m pole (average speed)

5 m pole			
bulbs	**HPS power consumption cost (BD)**	**LED System Cost (BD)**	**Cost saved (BD)**
1	102.312	28.67933	73.63267
2	204.624	57.35867	147.2653
4	409.248	114.7173	294.5307

Table 5-8: Cost saved by Intelligent LED system compared to HPS for 8m pole (average speed)

8 m pole			
bulbs	**HPS power consumption cost (BD)**	**LED System Cost (BD)**	**Cost saved (BD)**
1	365.4	102.4262	262.9738
2	730.8	204.8524	525.9476
4	1461.6	409.7048	1051.895

It’s clearly seen that the saved money has a big weight, so the intelligent system beats the real one by saving a huge amount of money.

Chapter 6
Conclusion

6.1 Conclusion

Finally, we have approached to propose a solution to an issue that is affecting each country's sources of electricity and developing an economic problem. As seen from our calculations, the contemporary lights gadget is consuming a lot of power and money whilst it can be much less through switching to an LED-based system, and even save greater through implementing an Intelligent LED System. After researching and getting the necessary data and information about the project, the main problem has been identified, hence the motivation of our project. After forming a solid picture of the issue, we have started looking for an answer. The aim was to build an intelligent system and develop a prototype that would simulate our idea's real-life scenario to make it something realistic, not just thoughts. By first choosing our controller, being Arduino, together with Java, we were able to design the prototype which made things much more dynamic. After finishing the design, an interface was also made so that the system created is also linked to the user interface to process data coming from Arduino and convert it to something that non-experts can easily use and handle. Then a comparison was made between these systems, HPS and Intelligent LED-based systems.

It was seen that the Intelligent LED system was the best solution for energy savings and costs, and what that meant was that our project objective was being met.

6.2 Future work

The completion of this project is the very first step in its development. As we look forward, this project will be increasingly enhanced in the future until it becomes a commercially viable product. And our future aspects for this project is:

- Reducing the wiring between each pole, using a wireless connection has to be made such as Bluetooth or Wi-Fi using Zigbee or other modules.
- Use solar panels so they can store energy and use the energy to supply the system which can further reduce the cost.
- Add power consumption directly to the interface so the interface can calculate the consumption automatically.

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Appendix A

Work (Gantt chart)

Task Activity	Timeline					
	Jan 16 – Feb 2	Feb 15 – Feb 27	Feb 28 – Mar 12	Mar 13 – Mar 16	Mar 17 – Apr 19	Apr 20 – May 13
Gathering information about the project						
Collect the main parts for the design (Arduino, sensors, etc.)						
Designing the prototype						
Programming Arduino & Java						
Creating the interface						
Troubleshooting						
Writing the report						

Appendix B

JavaScript code

```
var express = require('express'); // library
var app = express(); // create application from the library
var httpServer = require("http").createServer(app); // create server for the application
var five = require("johnny-five");  // IoT library to control the Arduino
var io = require('socket.io')(httpServer);  // socket to send messeges to the client
var moment = require("moment");

var port = 3000; // port used
app.use(express.static(__dirname + '/public'));
app.get('/', function (req, res) {
   res.sendFile(__dirname + '/public/index.html');
});

httpServer.listen(port);
console.log('Server available at http://localhost:' + port);

var ledPins = [8, 9, 10, 11];   // array of leds pin numbers in order
var motionPins = [2, 3, 4, 5];  // array of motions sensors pin numbers in order
var mode1Led;   // middle led

var leds = [];          // all led instances
var motions = [];       // all motion instances
var carPassed = [];     // number of car passed on each sensor night
var carPassedAll = [];     // number of car passed on each sensor all cars

var ldr;                // ldr sensor
var ldrPower;           // current reading of the ldt sensor
var ldrThreshold = 300; // when it should be dark

// Arduino board connection
var board = new five.Board();



var hwStatus = [];

board.on("ready", function () {
   console.log('Arduino connected');

   for (var i = 0; i < ledPins.length; i++) {
      leds.push(new five.Led(ledPins[i]));    // add leds
   }

   for (var i = 0; i < motionPins.length; i++) {
```

```
    motions.push(new five.Motion(motionPins[i]));   // add motions
    carPassed.push(0);  // intially all zeros
    carPassedAll.push(0);  // intially all zeros

    hwStatus.push({
        sensor: i,
        action: "",
        carPassed: 0,
        carPassedAll: 0,
         light: "",

        time: "
    });
}

mode1Led = new five.Led(22);    // connect middle led

ldr = new five.Sensor({ // connect ldr sensor
    pin: 'A0',
    freq: 300
});


function montionDetection(i, socket) {

    // when the motion start this will happen
    motions[i].on('motionstart', () => {
        // check if car passed less than the car passed for previous sensor
        if (i >= 1) {
            if (carPassed[i - 1] < (carPassed[i] + 1)) {
                console.log("CANT START");
                return; // no in the middle attack
            }
        }

        carPassedAll[i] += 1;

        console.log("A0: " + ldrPower);

        if (ldrPower > ldrThreshold) {

        hwStatus[i].action = "motionstart";
        hwStatus[i].carPassedAll = carPassedAll[i];
        hwStatus[i].time = moment().format('h:mm:ss a');

        socket.emit("sensorMotion", hwStatus);

            return;
        } else {
```

```
            //mode1Led.on();
        }

        console.log('motion START at PIR [', i + 1, "] Cars Passed [", carPassed[i] + 1, "] -> ",
moment().format('h:mm:ss a'));


        // send to the html to show a car entered
        if (i == 0) {
            socket.emit("motionStart"); // a car jsut entered the road
        }

        leds[i].on();

        hwStatus[i].action = "motionstart";
        hwStatus[i].carPassed = carPassed[i] + 1;
        hwStatus[i].carPassedAll = carPassedAll[i];
        hwStatus[i].light = 'ON';
        hwStatus[i].time = moment().format('h:mm:ss a');


        if (i + 1 < motionPins.length) {
            leds[i + 1].on();
             hwStatus[i + 1].light = 'ON';
        }




        carPassed[i] += 1;

        if (i >= 1) {
            if (carPassed[i - 1] <= carPassed[i]) {
                leds[i - 1].off();
                hwStatus[i - 1].light = 'OFF';

            }
        }
        // send to the html the motion
        socket.emit("sensorMotion", hwStatus);
    });
    // register the motion when end the below will happen
    // check if any motion from sensors not started from the beginning and ignore it
    // check the car passed with the sensor before
    // switch off the lights and send to the socket to show leds off
    motions[i].on('motionend', () => {

        if (i >= 1) {
```

```
            if (carPassed[i - 1] < (carPassed[i] + 1)) {
                console.log("CANT END");
            }

            //return; // no in the middle attack

            if (carPassed[i - 1] <= carPassed[i]) {
                leds[i - 1].off();

                hwStatus[i - 1].light = "OFF";

                //leds[i].off();
            }
        }

        console.log("A0: " + ldrPower);

        if (ldrPower > ldrThreshold) {
            // send to the html to show Reading
            socket.emit("ldrReading", "LIGHT");
            return;
        } else {
            // mode1Led.on();
        }


            hwStatus[i].action = "motion end";
            hwStatus[i].time = moment().format('h:mm:ss a');


        console.log('motion END at PIR [', i + 1, "] -> ", moment().format('h:mm:ss a'));

        if (i == motionPins.length - 1) {

            socket.emit("motionEnd"); // a car jsut finished the road


            if (carPassed[i - 1] == carPassed[i]) {
                leds[i].off();

                hwStatus[i].light = "OFF";

            }
        }

        socket.emit("sensorMotion", hwStatus);

    });
}
```

```
//Socket connection handler
io.on('connection', function (socket) {
   console.log(socket.id);

   for (i = 0; i < motionPins.length; i++) {
      montionDetection(i, socket);    // start detection on all sensors
   }


      // ldr reading
ldr.on("data", function (err, data) {

   if (ldrPower != this.value) {

   console.log('data: ', this.value);

      var reading = '';
      if (this.value < ldrThreshold) {
         reading = 'DARK';
      } else {
         reading = 'LIGHT';
      }
       // send the ldr reading to the html
      socket.emit("ldrReading", reading);

   }
   ldrPower = this.value;
   //console.log('A0: ' + ldrPower);
   if (ldrPower <= ldrThreshold) {
      mode1Led.on();  // open middle lights if it is dark
   } else {
      // if it is not dark; turn all the lights off
      mode1Led.off();
      for (i = 0; i < leds.length; i++) {
         leds[i].off();
      }
       socket.emit("ldrReading", "LIGHT");

       //

         for (var i = 0; i < motionPins.length; i++) {

      hwStatus[i].light = '';


      socket.emit("sensorMotion", hwStatus);

      }
```

```
        //
    }
  });

socket.on('led:on', function (data) {

        for (var x = 0; x < ledPins.length; x++) {
          leds[x].on();

        }

        console.log('LED ON RECEIVED');
    });
    socket.on('led1:on', function (data) {
          leds[0].on();


        console.log('LED 1 ON RECEIVED');
    });


     socket.on('led:off', function (data) {

        for (var x = 0; x < ledPins.length; x++) {
          leds[x].off();
        }
        console.log('LED OFF RECEIVED');
    });

    socket.on('led1:off', function (data) {
          leds[0].off();

        console.log('LED 1 OFF RECEIVED');
    });

 socket.on('led2:off', function (data) {
          leds[1].off();

        console.log('LED 2 OFF RECEIVED');
    });
     socket.on('led3:off', function (data) {
          leds[2].off();

        console.log('LED 3 OFF RECEIVED');
    });
     socket.on('led4:off', function (data) {
          leds[3].off();
```

```
            console.log('LED 4 OFF RECEIVED');
        });
        // when reset button clicked to switch off all the lights
        socket.on('reset', function (data) {

            for (var i = 0; i < motionPins.length; i++) {
                carPassed[i] = 0;
                carPassedAll[i] = 0;

                leds[i].off();

            hwStatus[i].action = "";
            hwStatus[i].carPassed = 0;
            hwStatus[i].carPassedAll = 0;
            hwStatus[i].light = '';
            hwStatus[i].time = '';

            socket.emit("sensorMotion", hwStatus);

            }
        });
    });
});
```

Appendix C

Cascading Style Sheets code

```
.ON {
   background-image: -webkit-linear-gradient(top, #f9a004 0%, #e0ac45 50%, #ead698 100%);
}
.OFF {
   background-image: -webkit-linear-gradient(top, #d3d3d3 0%, #d3d3d3 0%, #d3d3d3 0%);
}
.SENON {
   background-image: -webkit-linear-gradient(top, #fb1304 0%, #e35843 50%, #edad99 100%);
}
.led {
   border-radius: 15px;
   width: 30px;
   height: 30px;
   box-shadow: 0px 0px 3px black;
   margin: 5px;
   /*zoom: 5;*/
}
.snesor {
   border-radius: 5px;
   width: 30px;
   height: 30px;
   box-shadow: 0px 0px 3px black;
   margin: 5px;
   /*zoom: 5;*/
}
.led:after {
   display: block;
   content: '';
   margin-left: 1px;
   margin-right: 1px;
```

```
    width: 8px;
    height: 6px;
    -webkit-border-top-right-radius: 4px 3px;
    -webkit-border-top-left-radius: 4px 3px;
    -webkit-border-bottom-right-radius: 4px 3px;
    -webkit-border-bottom-left-radius: 4px 3px;
    background-image: -webkit-linear-gradient(top, rgba(255,255,255,0.8) 0%, rgba(255,255,255,0.2) 100%);
}
```

Appendix D

HyperText Markup Language code and interface flowchart

```
<!DOCTYPE html>
<html lang="en" ng-app="myApp" class="no-js"> <!--<![endif]-->
<head>
  <meta charset="utf-8">
  <meta http-equiv="X-UA-Compatible" content="IE=edge">
  <title>Smart Traffic</title>

    <link rel="stylesheet" href="css/vendor/bootstrap.css">
    <link rel="stylesheet" href="css/main.css">
    <link rel="stylesheet" href="css/custom.css">

</head>
<body ng-controller="ArduController">
  <script src="lib/socket.io-client/dist/socket.io.js"></script>
  <script src="lib/angular/angular.js"></script>
  <script src="lib/angular-socket-io/socket.js"></script>
   <script src="js/index.js"></script>

<div class="container" >
  <div class="panel">
    <div class="panel-body">

<h1>{{"Led controller for first street"}}</h1>
  <div >
    <button ng-click="ledOn()">OnALL</button>
    <button ng-click="ledOff()">OffALL</button>
    <button ng-click="led1On()">On1</button>
    <button ng-click="led1Off()">Off1</button>
     <button ng-click="led2On()">On2</button>
    <button ng-click="led2Off()">Off2</button>
     <button ng-click="led3On()">On3</button>
    <button ng-click="led3Off()">Off3</button>
     <button ng-click="led4On()">On4</button>
    <button ng-click="led4Off()">Off4</button>
    <button ng-click="reset()">Reset</button>

        <h1>Cars entered road: {{carsCount}}</h1>
        <h1>Cars finished road: {{carsCountEnd}}</h1>

        <h1> LDR Reading: {{ldrReading}}</h1>

  </div>
```

```
  </div>
 </div>


<div class="panel">
   <div class="panel-body">
   <legend>Sensor Status</legend>

 <table class="table table-hover table-striped">
   <thead>
   <tr>
    <th class="">Sensor </th>
    <th class=""> Action</th>
    <th class="">Car paased night</th>
    <th class="">Car passed all</th>
    <th class=""> LED</th>
    <th class=""> Time</th>

   </tr>
   </thead>

<tbody ng-repeat="senesor in senesors">
       <tr>
          <td>{{senesor.sensor + 1}}</td>
          <td>{{senesor.action}}</td>
          <td>{{senesor.carPassed}}</td>
          <td>{{senesor.carPassedAll}}</td>
          <td>{{senesor.light}}</td>
          <td>{{senesor.time}}</td>

       </tr>
   </tbody>

   </tbody>

   </table>

   </div>

</div>

<div class="panel">

   <div class="panel-body">
   <legend>Sensor Interface</legend>

        <h1 align="center">START</h1>
```

```
 <table class="table" >

  <tbody ng-repeat="senesor in senesors">

      <tr>
        <td><div class="{{senesor.light}} led"></div></td>
        <td><div class="{{ldrReading == 'DARK' ? 'ON' : 'OFF'}} led"></div></td>
        <td><div class="{{senesor.light}} led"></div></td>

        <td rowspan="2"><img src="road.png" width='50px' height="150px"></td>

        <td><div class="{{senesor.light}} led"></div></td>
        <td><div class="{{ldrReading == 'DARK' ? 'ON' : 'OFF'}} led"></div></td>
        <td><div class="{{senesor.light}} led"></div></td>
      </tr>

      <tr>
        <td align="right" colspan="3"><div class="{{senesor.action == 'motionstart' ?
'SENON' : 'OFF'}} snesor"></div></td>
      </tr>

   </tbody>

   </table>

       <h1 align="center">END</h1>

   </div>

</div>

 </div>

var app = angular.module('myApp', ['btford.socket-io']).
   factory('mySocket', function (socketFactory) {
     return socketFactory();
   }).
   controller('ArduController', function ($scope,mySocket) {

     $scope.carsCount = 0;
     $scope.carsCountEnd = 0;

     var updateCounter = function() {
       $scope.carsCount = $scope.carsCount + 1;
     };


     var updateCounterEnd = function() {
       $scope.carsCountEnd = $scope.carsCountEnd + 1;
```

```
};

$scope.ledOn = function () {

   mySocket.emit('led:on');
   console.log('LED ON');
};

 $scope.led1On = function () {

   mySocket.emit('led1:on');
   console.log('LED 1 ON');
};

$scope.led2On = function () {

   mySocket.emit('led2:on');
   console.log('LED 2 ON');
};

$scope.led3On = function () {

   mySocket.emit('led3:on');
   console.log('LED 3 ON');
};

$scope.led4On = function () {

   mySocket.emit('led4:on');
   console.log('LED 4 ON');
};

$scope.ledOff = function () {

   mySocket.emit('led:off');
   console.log('LED OFF');
};

 $scope.led1Off = function () {

   mySocket.emit('led1:off');
   console.log('LED 1 OFF');
};

 $scope.led2Off = function () {

   mySocket.emit('led2:off');
   console.log('LED 2 OFF');
```

```
        };

        $scope.led3Off = function () {

            mySocket.emit('led3:off');
            console.log('LED 3 OFF');
        };

        $scope.led4Off = function () {

            mySocket.emit('led4:off');
            console.log('LED 4 OFF');
        };

$scope.reset = function () {

            mySocket.emit('reset');

            $scope.carsCount = 0;
            $scope.carsCountEnd = 0;
        };
$scope.senesors = [];
        mySocket.on("sensorMotion", function(data) {
            // json

            $scope.senesors = data;

            console.log(data);

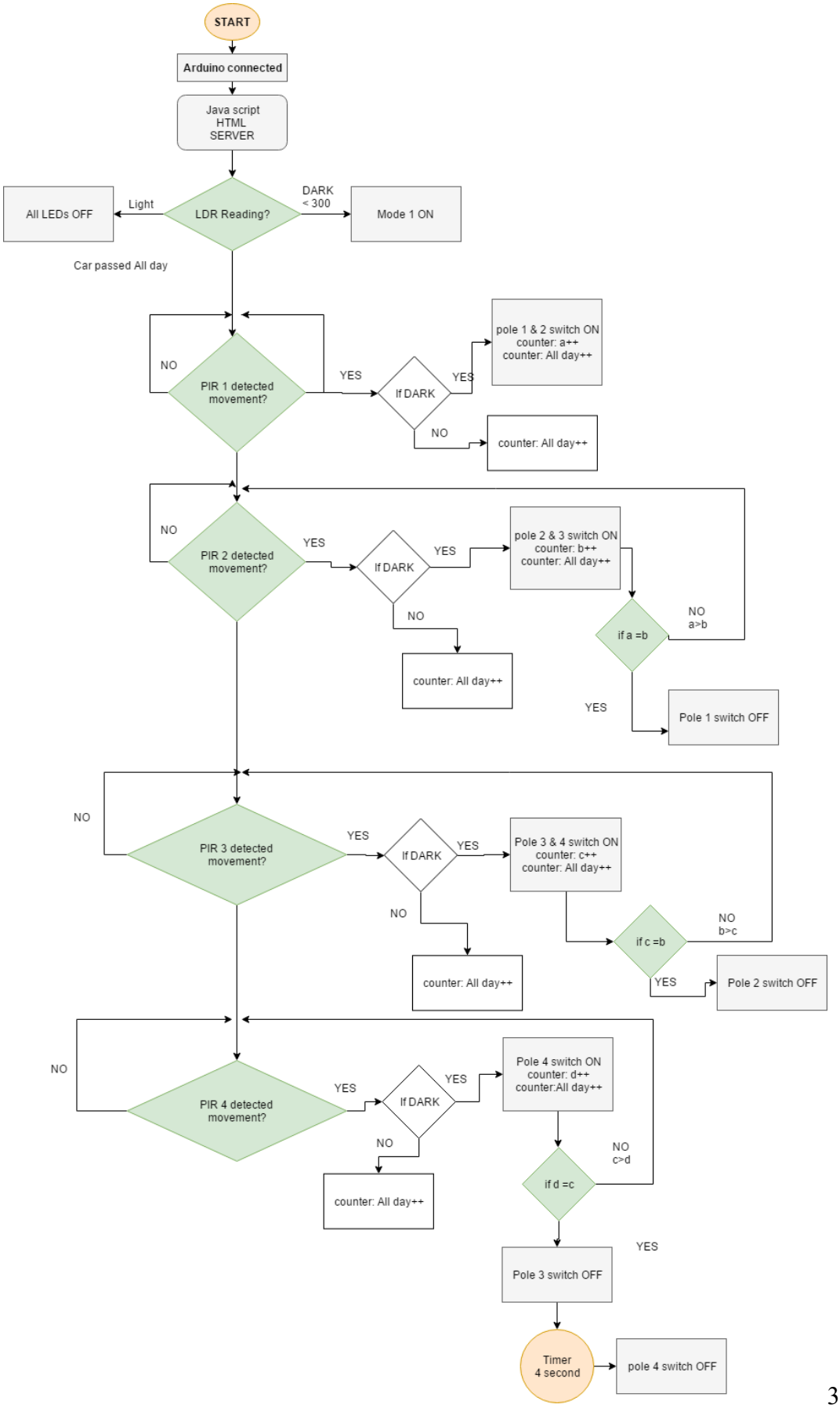

        });

        $scope.ldrReading = '';

        mySocket.on("ldrReading", function(data) {
            // json

            $scope.ldrReading = data;


        });
```


START
Arduino connected
Java script
HTML
SERVER
LDR Reading?
Light
All LEDs OFF
DARK
< 300
Mode 1 ON
Car passed All day
PIR 1 detected movement?
NO
YES
If DARK
YES
pole 1 & 2 switch ON
counter: a++
counter: All day++
NO
counter: All day++
PIR 2 detected movement?
NO
YES
If DARK
YES
pole 2 & 3 switch ON
counter: b++
counter: All day++
NO
counter: All day++
if a =b
NO
a>b
YES
Pole 1 switch OFF
PIR 3 detected movement?
NO
YES
If DARK
YES
Pole 3 & 4 switch ON
counter: c++
counter: All day++
NO
counter: All day++
if c =b
NO
b>c
YES
Pole 2 switch OFF
PIR 4 detected movement?
NO
YES
If DARK
YES
Pole 4 switch ON
counter: d++
counter:All day++
NO
counter: All day++
if d =c
NO
c>d
YES
Pole 3 switch OFF
Timer
4 second
pole 4 switch OFF

3

Appendix E

IRF840 MOSFET DATASHEET

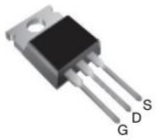
VISHAY.
www.vishay.com

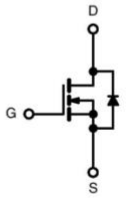
IRF840, SiHF840
Vishay Siliconix

Power MOSFET

PRODUCT SUMMARY		
V_{DS} (V)	500	
$R_{DS(on)}$ (Ω)	V_{GS} = 10 V	0.85
Q_g max. (nC)	63	
Q_{gs} (nC)	9.3	
Q_{gd} (nC)	32	
Configuration	Single	

TO-220AB





N-Channel MOSFET

FEATURES

- Dynamic dV/dt rating
- Repetitive avalanche rated
- Fast switching
- Ease of paralleling
- Simple drive requirements
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

Available
RoHS*
Available

Note
* This datasheet provides information about parts that are RoHS-compliant and / or parts that are non-RoHS-compliant. For example, parts with lead (Pb) terminations are not RoHS-compliant. Please see the information / tables in this datasheet for details.

DESCRIPTION

Third generation power MOSFETs from Vishay provide the designer with the best combination of fast switching, ruggedized device design, low on-resistance and cost-effectiveness.
The TO-220AB package is universally preferred for all commercial-industrial applications at power dissipation levels to approximately 50 W. The low thermal resistance and low package cost of the TO-220AB contribute to its wide acceptance throughout the industry.

ORDERING INFORMATION	
Package	TO-220AB
Lead (Pb)-free	IRF840PbF
	SiHF840-E3
SnPb	IRF840
	SiHF840

ABSOLUTE MAXIMUM RATINGS (T_C = 25 °C, unless otherwise noted)

PARAMETER			SYMBOL	LIMIT	UNIT
Drain-Source Voltage			V_{DS}	500	V
Gate-Source Voltage			V_{GS}	± 20	V
Continuous Drain Current	V_{GS} at 10 V	T_C = 25 °C	I_D	8.0	A
		T_C = 100 °C		5.1	
Pulsed Drain Current [a]			I_{DM}	32	
Linear Derating Factor				1.0	W/°C
Single Pulse Avalanche Energy [b]			E_{AS}	510	mJ
Repetitive Avalanche Current [a]			I_{AR}	8.0	A
Repetitive Avalanche Energy [a]			E_{AR}	13	mJ
Maximum Power Dissipation	T_C = 25 °C		P_D	125	W
Peak Diode Recovery dV/dt [c]			dV/dt	3.5	V/ns
Operating Junction and Storage Temperature Range			T_J, T_{stg}	-55 to +150	°C
Soldering Recommendations (Peak temperature) [d]	for 10 s			300	
Mounting Torque	6-32 or M3 screw			10	lbf · in
				1.1	N · m

Notes
a. Repetitive rating; pulse width limited by maximum junction temperature (see fig. 11).
b. V_{DD} = 50 V, starting T_J = 25 °C, L = 14 mH, R_g = 25 Ω, I_{AS} = 8.0 A (see fig. 12).
c. I_{SD} ≤ 8.0 A, dI/dt ≤ 100 A/µs, V_{DD} ≤ V_{DS}, T_J ≤ 150 °C.
d. 1.6 mm from case.

For technical questions, contact: hvm@vishay.com
THIS DOCUMENT IS SUBJECT TO CHANGE WITHOUT NOTICE. THE PRODUCTS DESCRIBED HEREIN AND THIS DOCUMENT ARE SUBJECT TO SPECIFIC DISCLAIMERS, SET FORTH AT www.vishay.com/doc?91000

THERMAL RESISTANCE RATINGS				
PARAMETER	SYMBOL	TYP.	MAX.	UNIT
Maximum Junction-to-Ambient	R_{thJA}	-	62	°C/W
Case-to-Sink, Flat, Greased Surface	R_{thCS}	0.50	-	
Maximum Junction-to-Case (Drain)	R_{thJC}	-	1.0	

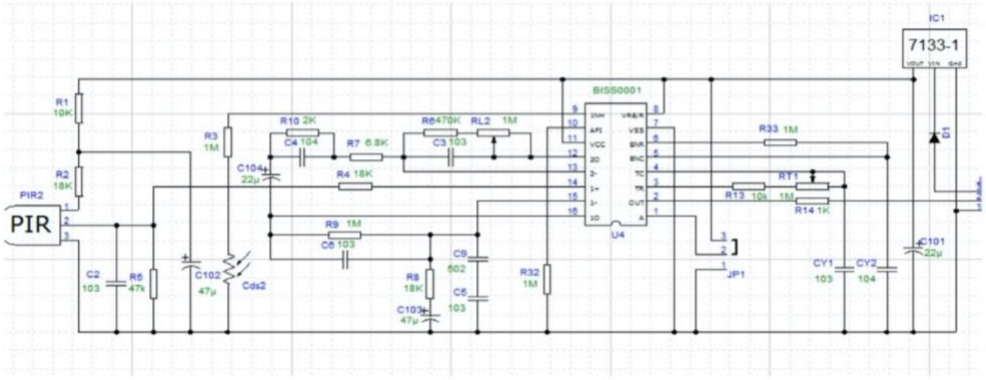
SPECIFICATIONS (T_J = 25 °C, unless otherwise noted)						
PARAMETER	SYMBOL	TEST CONDITIONS	MIN.	TYP.	MAX.	UNIT
Static						
Drain-Source Breakdown Voltage	V_{DS}	V_{GS} = 0 V, I_D = 250 µA	500	-	-	V
V_{DS} Temperature Coefficient	$\Delta V_{DS}/T_J$	Reference to 25 °C, I_D = 1 mA	-	0.78	-	V/°C
Gate-Source Threshold Voltage	$V_{GS(th)}$	V_{DS} = V_{GS}, I_D = 250 µA	2.0	-	4.0	V
Gate-Source Leakage	I_{GSS}	V_{GS} = ± 20 V	-	-	± 100	nA
Zero Gate Voltage Drain Current	I_{DSS}	V_{DS} = 500 V, V_{GS} = 0 V	-	-	25	µA
		V_{DS} = 400 V, V_{GS} = 0 V, T_J = 125 °C	-	-	250	
Drain-Source On-State Resistance	$R_{DS(on)}$	V_{GS} = 10 V, I_D = 4.8 A [b]	-	-	0.85	Ω
Forward Transconductance	g_{fs}	V_{DS} = 50 V, I_D = 4.8 A [b]	4.9	-	-	S
Dynamic						
Input Capacitance	C_{iss}	V_{GS} = 0 V, V_{DS} = 25 V, f = 1.0 MHz, see fig. 5	-	1300	-	pF
Output Capacitance	C_{oss}		-	310	-	
Reverse Transfer Capacitance	C_{rss}		-	120	-	
Total Gate Charge	Q_g	V_{GS} = 10 V, I_D = 8 A, V_{DS} = 400 V, see fig. 6 and 13 [b]	-	-	63	nC
Gate-Source Charge	Q_{gs}		-	-	9.3	
Gate-Drain Charge	Q_{gd}		-	-	32	
Turn-On Delay Time	$t_{d(on)}$	V_{DD} = 250 V, I_D = 8 A, R_g = 9.1 Ω, R_D = 31 Ω, see fig. 10 [b]	-	14	-	ns
Rise Time	t_r		-	23	-	
Turn-Off Delay Time	$t_{d(off)}$		-	49	-	
Fall Time	t_f		-	20	-	
Internal Drain Inductance	L_D	Between lead, 6 mm (0.25") from package and center of die contact 	-	4.5	-	nH
Internal Source Inductance	L_S		-	7.5	-	
Gate Input Resistance	R_g	f = 1 MHz, open drain	0.6	-	2.8	Ω
Drain-Source Body Diode Characteristics						
Continuous Source-Drain Diode Current	I_S	MOSFET symbol showing the integral reverse p - n junction diode 	-	-	8.0	A
Pulsed Diode Forward Current [a]	I_{SM}		-	-	32	
Body Diode Voltage	V_{SD}	T_J = 25 °C, I_S = 8 A, V_{GS} = 0 V [b]	-	-	2.0	V
Body Diode Reverse Recovery Time	t_{rr}	T_J = 25 °C, I_F = 8 A, dI/dt = 100 A/µs [b]	-	460	970	ns
Body Diode Reverse Recovery Charge	Q_{rr}		-	4.2	8.9	µC
Forward Turn-On Time	t_{on}	Intrinsic turn-on time is negligible (turn-on is dominated by L_S and L_D)				

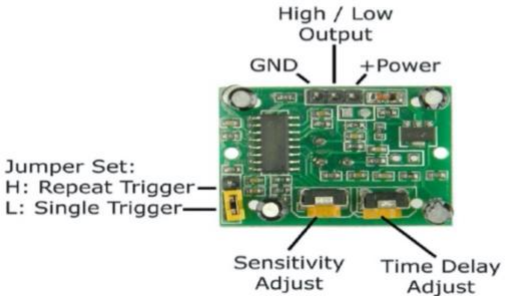
Notes

a. Repetitive rating; pulse width limited by maximum junction temperature (see fig. 11).
b. Pulse width ≤ 300 µs; duty cycle ≤ 2 %.

Appendix F

HC-SR501 PIR MOTION DETECTOR





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1 working voltage range :DC 4.5-20V

2 Quiescent Current :50uA

3 high output level 3.3 V / Low 0V

4. Trigger L trigger can not be repeated / H repeated trigger

5. circuit board dimensions :32 * 24 mm

6. maximum 110 ° angle sensor

7. 7 m maximum sensing distance

Product Type	HC--SR501 Body Sensor Module
Operating Voltage Range	5-20VDC
Quiescent Current	<50uA
Level output	High 3.3 V /Low 0V
Trigger	L can not be repeated trigger/H can be repeated trigger(Default repeated trigger)
Delay time	5-300S(adjustable) Range (approximately .3Sec -5Min)
Block time	2.5S(default)Can be made a range(0.xx to tens of seconds
Board Dimensions	32mm*24mm
Angle Sensor	<110 ° cone angle
Operation Temp.	-15-+70 degrees
Lens size sensor	Diameter:23mm(Default)

Appendix G:

Arduino MEGA 2560 Datasheet

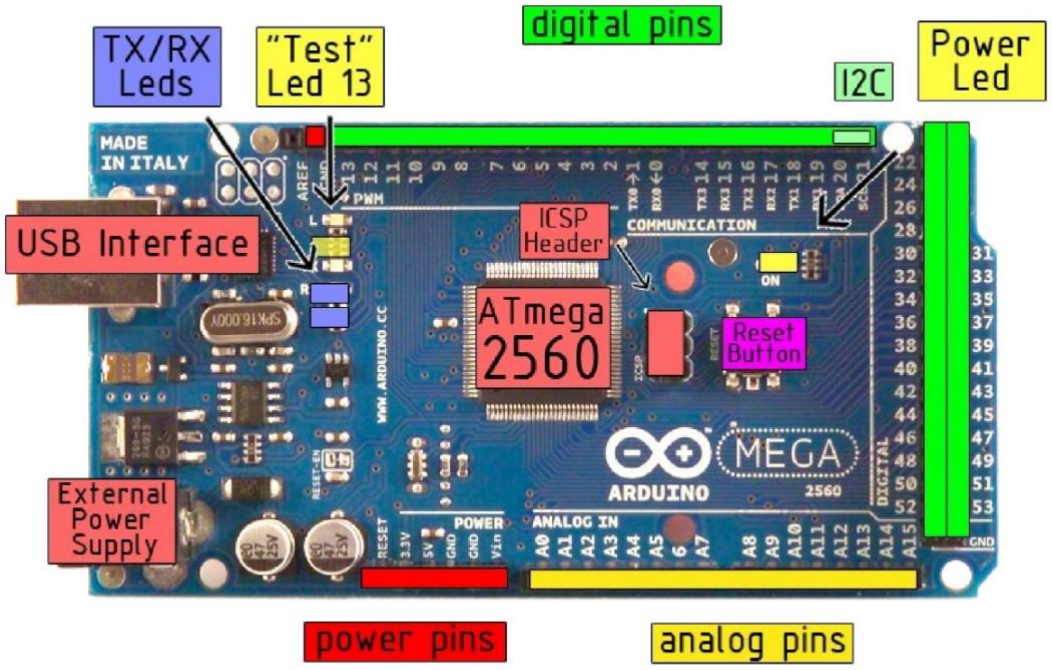
Technical Specification

EAGLE files: arduino-mega2560-reference-design.zip Schematic: arduino-mega2560-schematic.pdf

Summary

Microcontroller	ATmega2560
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limits)	6-20V
Digital I/O Pins	54 (of which 14 provide PWM output)
Analog Input Pins	16
DC Current per I/O Pin	40 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	256 KB of which 8 KB used by bootloader
SRAM	8 KB
EEPROM	4 KB
Clock Speed	16 MHz

the board



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ملخص المشروع

في مملكة البحرين يتم استخدام طرق للتحكم في إنارة مصابيح الطرق بواسطة الوقت واستشعار الضوء، هذه الطرق غير كافية تماما لتحقيق دقة أفضل في التحكم بسبب غياب تقنية استشعار الحركة للسيارات أو الأشخاص ضمن منظومة التحكم بإنارة المصابيح، وهذا ما يدفع المهندسين لدراسة وابتكار طرق وحلول جديدة لتحسين أداء الإنارة وتلبية الضرورة فيها بتقليل استهلاك الطاقة وبكلفة أقل.

تتمثل فكرة هذا المشروع في بناء نظام ديناميكي ذكي للتحكم بالمصابيح يعمل على تحسين عملها الحالي في حالات، إما تضيء أو لا تضيء أو إضاءة خافتة دون إحداث تغييرات على عملها وتلبية الحاجة الضرورية منها، وذلك حل لعدم استنفاذ وإهدار موارد الطاقة فيما لا يجدي.

تمنع هذه الفكرة العمل الغير ضروري للأضواء، وذلك عند عدم وجود مركبات أو أشخاص يمشون في منطقة عملها. يستعمل المشروع نظام تحكم Arduino بمساعدة جهاز مستشعر للحركة لإعطاء الأضواء إشارة بالتشغيل بناءًا على حركة المرور. إضافة إلى ذلك تم ربط واجهة المستخدم UI مباشرة بنظام المراقبة ورصد الحركة المرورية للتحكم.

لا يضمن تطبيق نظام إضاءة الشوارع الذكي توفير الطاقة فحسب، بل يمكن أن يساعد أيضًا على إطالة وقت خدمة الإضاءة نفسها نظرًا لعدم تشغيلها بشكل دائم. علاوة على ذلك، فإنها توفر نفقات عمل الصيانة والمواد بشكل كبير.